

SAVEETHA SCHOOL OF ENGINEERING
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CSA09 –JAVA PROGRAMMING

1. Write a program to count all the prime and composite numbers entered by the user.

Sample Input:

Enter the numbers

4
54
29
71
7
59
98
23

Sample Output:

Composite number:3

Prime number:5

Test cases:

1. 33, 41, 52, 61, 73, 90
2. TEN, FIFTY, SIXTY-ONE, SEVENTY-SEVEN, NINE
3. 45, 87, 09, 5.0, 2.3, 0.4
4. -54, -76, -97, -23, -33, -98
5. 45, 73, 00, 50, 67, 44

The screenshot shows a Java IDE with a file named 'ain.java'. The code is as follows:

```
import java.util.Scanner;
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int size = sc.nextInt();

        int arr[] = new int[size];

        System.out.println("Enter elements:");

        for(int i=0;i<size;i++)
            arr[i]=sc.nextInt();

        System.out.print("Enter M: ");
        int m=sc.nextInt();

        System.out.print("Enter N: ");
        int n=sc.nextInt();

        Arrays.sort(arr);
```

The output window on the right shows the following execution:

```
Enter array size: 5
Enter elements:
1 2 3 5 4
Enter M: 3
Enter N: 4
3th Maximum Number = 3
4th Minimum Number = 4
Sum = 7
Difference = -1

=== Code Execution Successful ===
```

2. Find the M^{th} maximum number and N^{th} minimum number in an array and then find the sum of it and difference of it.

Sample Input:

Array of elements = {14, 16, 87, 36, 25, 89, 34}

M = 1

N = 3

Sample Output:

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1stMaximum Number = 89

3rdMinimum Number = 25

Sum = 114

Difference = 64

Test cases:

1. {16, 16, 16 16, 16}, M = 0, N = 1
2. {0, 0, 0, 0}, M = 1, N = 2
3. {-12, -78, -35, -42, -85}, M = 3 , N = 3
4. {15, 19, 34, 56, 12}, M = 6 , N = 3
5. {85, 45, 65, 75, 95}, M = 5 , N = 7

<pre>1 import java.util.Scanner; 2 3 public class Main { 4 public static void main(String args[]) { 5 6 Scanner sc = new Scanner(System.in); 7 8 int total = 0; 9 10 for(int i=1;i<=4;i++) 11 { 12 System.out.print("Enter Denomination: "); 13 int denomination = sc.nextInt(); 14 15 System.out.print("Enter Number of Notes: "); 16 int notes = sc.nextInt(); 17 18 total = total + (denomination * notes); 19 } 20 21 System.out.println("Total Available Balance in ATM: " 22 + total); 23 }</pre>	<pre>Enter Denomination: 5 Enter Number of Notes: 3 Enter Denomination: 6 Enter Number of Notes: 2 Enter Denomination: 6 Enter Number of Notes: 8 Enter Denomination: 4 Enter Number of Notes: 8 Total Available Balance in ATM: 107 === Code Execution Successful ===</pre>
---	---

3. Write a program to print the total amount available in the ATM machine with the conditions applied.

Total denominations are 2000, 500, 200, 100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Sample Input:

Enter the 1st Denomination: 500

Enter the 1st Denomination number of notes: 4

Enter the 2nd Denomination: 100

Enter the 2nd Denomination number of notes: 20

Enter the 3rd Denomination: 200

Enter the 3rd Denomination number of notes: 32

Enter the 4th Denomination: 2000

Enter the 4th Denomination number of notes: 1

Sample Output:

Total Available Balance in ATM: 12400

Test Cases:

3 Hidden Test cases (Think Accordingly based on Denominations)

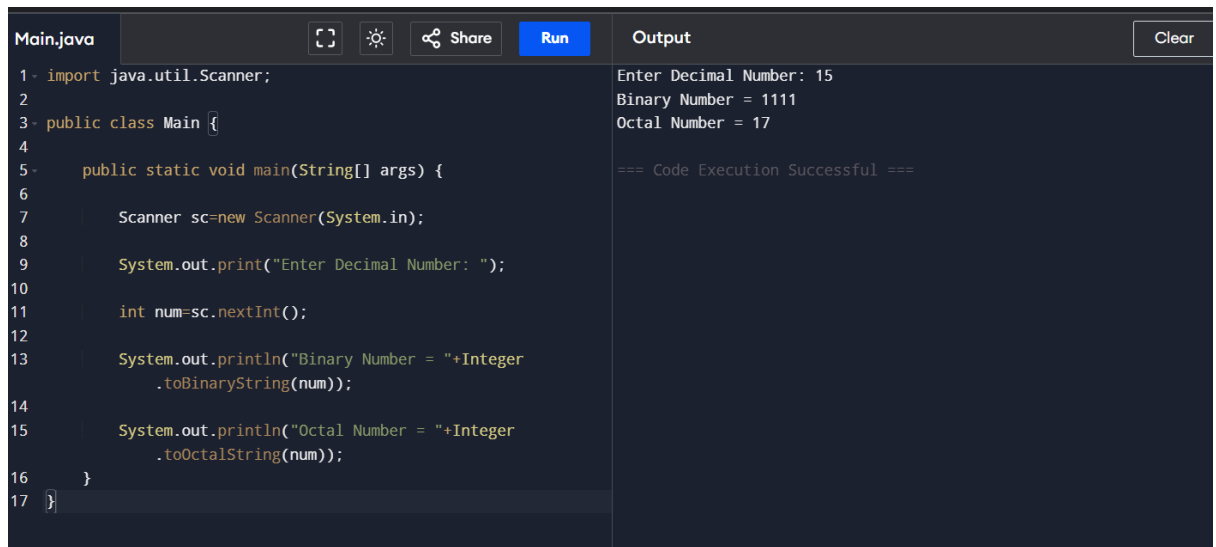
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```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("1.String Palindrome");  
        System.out.println("2.Number Palindrome");  
  
        int choice = sc.nextInt();  
  
        if(choice==1)  
        {  
            String str=sc.next();  
            String rev="";  
  
            for(int i=str.length()-1;i>=0;i--)  
                rev=rev+str.charAt(i);  
  
            if(str.equals(rev))  
                System.out.println("Palindrome");  
            else  
                System.out.println("Not Palindrome");  
        }  
    }  
}
```

1.String Palindrome
2.Number Palindrome

4. Write a program using choice to check
Case 1: Given string is palindrome or not
Case 2: Given number is palindrome or not
Sample Input:
Case = 1
String = MADAM
Sample Output:
Palindrome
Test cases:
1. MONEY
2. 5678765
3. MALAY12321ALAM
4. MALAYALAM
5. 1234.4321

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```
Main.java
1- import java.util.Scanner;
2
3- public class Main {
4
5-     public static void main(String[] args) {
6
7         Scanner sc=new Scanner(System.in);
8
9         System.out.print("Enter Decimal Number: ");
10
11         int num=sc.nextInt();
12
13         System.out.println("Binary Number = "+Integer
14             .toBinaryString(num));
15
16         System.out.println("Octal Number = "+Integer
17             .toOctalString(num));
18     }
19 }
```

Output

```
Enter Decimal Number: 15
Binary Number = 1111
Octal Number = 17

=== Code Execution Successful ===
```

5. Write a program to convert Decimal number equivalent to Binary number and octal numbers?

Sample Input:

Decimal Number: 15

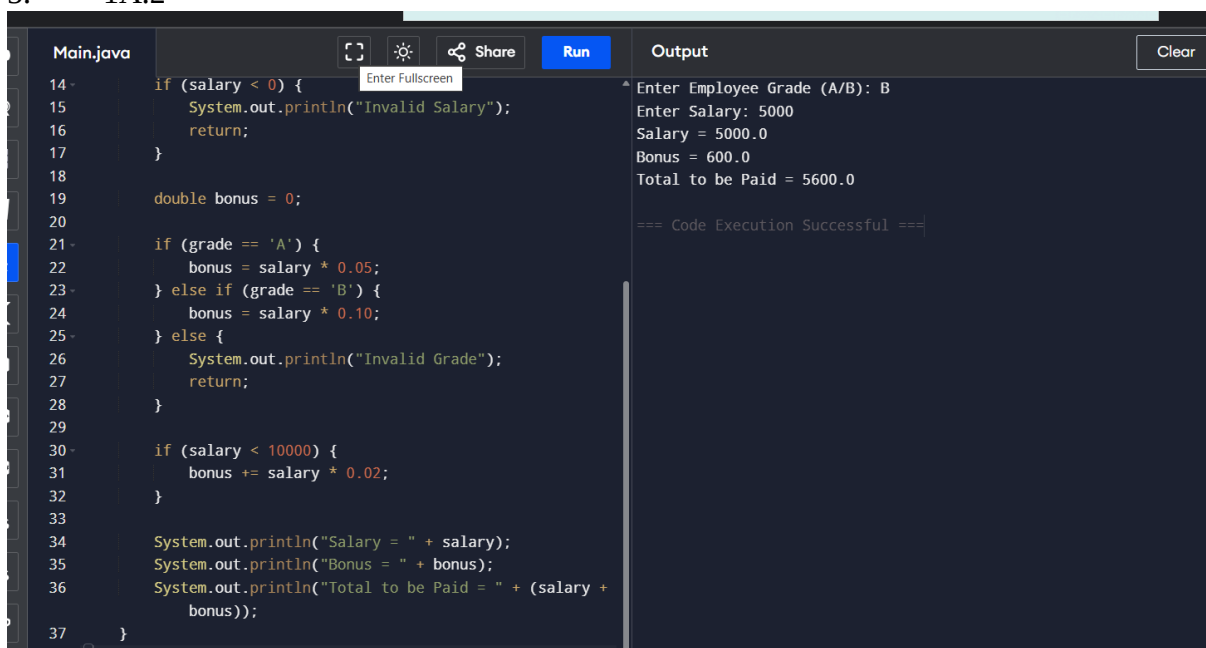
Sample Output:

Binary Number = 1111

Octal = 17

Test cases:

1. 111
2. 15.2
3. 0
4. B12
5. 1A.2



```
Main.java
14- if (salary < 0) {
15-     System.out.println("Invalid Salary");
16-     return;
17- }
18
19- double bonus = 0;
20
21- if (grade == 'A') {
22-     bonus = salary * 0.05;
23- } else if (grade == 'B') {
24-     bonus = salary * 0.10;
25- } else {
26-     System.out.println("Invalid Grade");
27-     return;
28- }
29
30- if (salary < 10000) {
31-     bonus += salary * 0.02;
32- }
33
34- System.out.println("Salary = " + salary);
35- System.out.println("Bonus = " + bonus);
36- System.out.println("Total to be Paid = " + (salary +
37-     bonus));
38- }
```

Output

```
Enter Employee Grade (A/B): B
Enter Salary: 5000
Salary = 5000.0
Bonus = 600.0
Total to be Paid = 5600.0

=== Code Execution Successful ===
```

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6. In an organization they decide to give bonus to all the employees on New Year. A 5% bonus on salary is given to the grade A workers and 10% bonus on salary to the grade B workers. Write a program to enter the salary and grade of the employee. If the salary of the employee is less than \$10,000 then the employee gets an extra 2% bonus on salary Calculate the bonus that has to be given to the employee and print the salary that the employee will get.

Sample Input & Output:

Enter the grade of the employee: B

Enter the employee salary: 50000

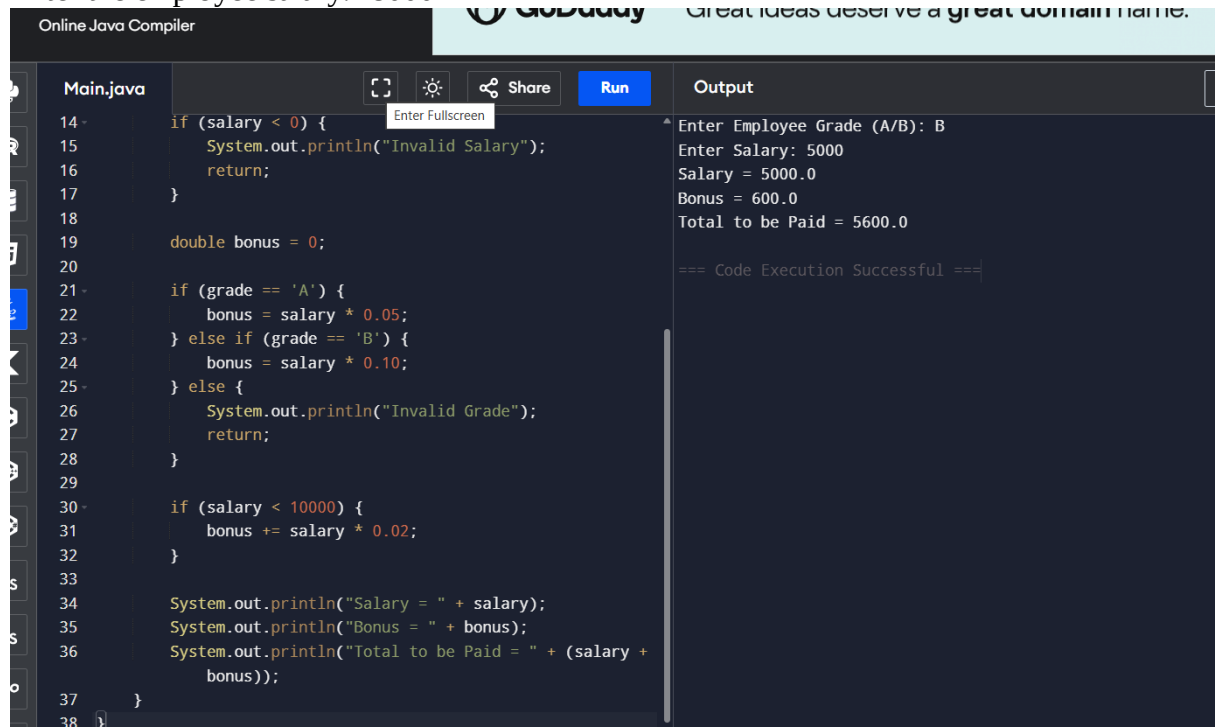
Salary=50000

Bonus=5000.0

Total to be paid:55000.0

Test cases:

1. Enter the grade of the employee: A
Enter the employee salary: 8000
2. Enter the grade of the employee: C
Enter the employee salary: 60000
3. Enter the grade of the employee: B
Enter the employee salary: 0
4. Enter the grade of the employee: 38000
Enter the employee salary: A
5. Enter the grade of the employee: B
Enter the employee salary: -8000



The screenshot shows an online Java compiler interface. On the left, the code for 'Main.java' is displayed, featuring logic for grade validation, bonus calculation (5% for 'A', 10% for 'B', and an additional 2% for salaries below 10,000), and final salary/bonus printing. On the right, the 'Output' pane shows the execution results for the input 'B' and salary '5000', resulting in a total payment of 5600.0. The status bar at the bottom indicates 'Code Execution Successful'.

```
Online Java Compiler

Main.java
14 if (salary < 0) {
15     System.out.println("Invalid Salary");
16     return;
17 }
18
19 double bonus = 0;
20
21 if (grade == 'A') {
22     bonus = salary * 0.05;
23 } else if (grade == 'B') {
24     bonus = salary * 0.10;
25 } else {
26     System.out.println("Invalid Grade");
27     return;
28 }
29
30 if (salary < 10000) {
31     bonus += salary * 0.02;
32 }
33
34 System.out.println("Salary = " + salary);
35 System.out.println("Bonus = " + bonus);
36 System.out.println("Total to be Paid = " + (salary +
37     bonus));
38 }
```

Output

```
Enter Employee Grade (A/B): B
Enter Salary: 5000
Salary = 5000.0
Bonus = 600.0
Total to be Paid = 5600.0

=== Code Execution Successful ===
```

7. Write a program to print the first n perfect numbers. (Hint Perfect number means a positive integer that is equal to the sum of its proper divisors)

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Sample Input:

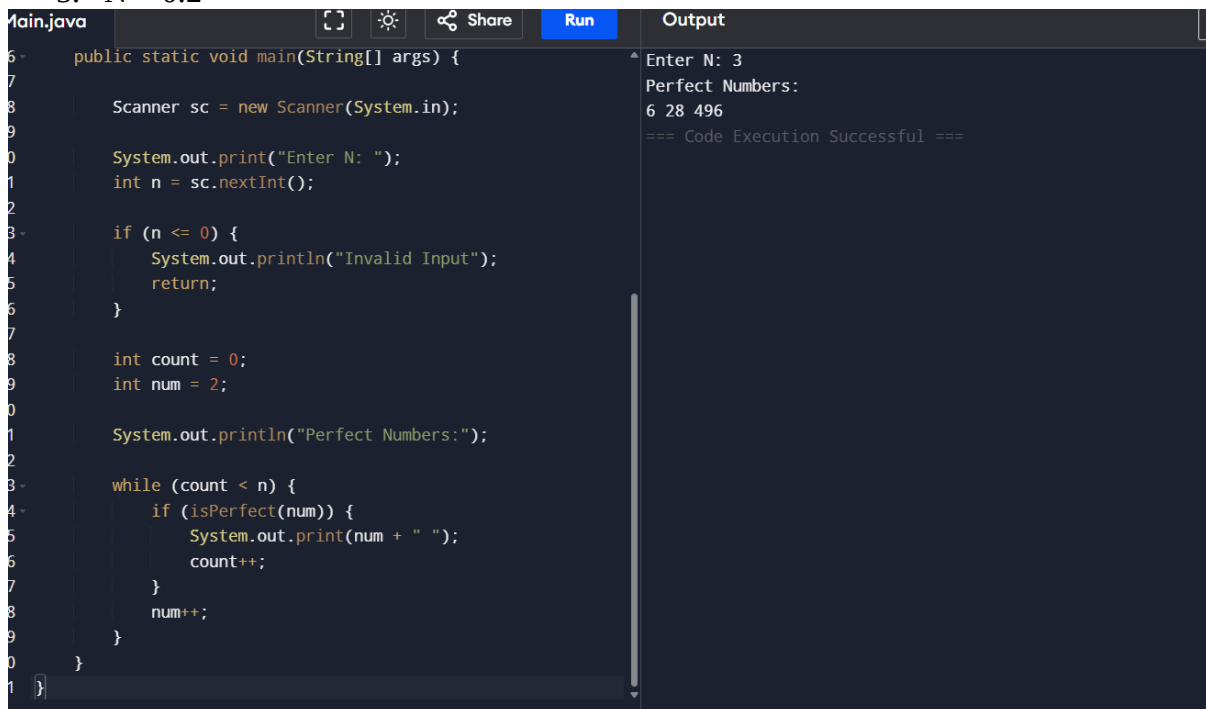
N = 3

Sample Output:

First 3 perfect numbers are: 6 , 28 , 496

Test Cases:

1. N = 0
2. N = 5
3. N = -2
4. N = -5
5. N = 0.2



```
Main.java
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter N: ");
    int n = sc.nextInt();

    if (n <= 0) {
        System.out.println("Invalid Input");
        return;
    }

    int count = 0;
    int num = 2;

    System.out.println("Perfect Numbers:");

    while (count < n) {
        if (isPerfect(num)) {
            System.out.print(num + " ");
            count++;
        }
        num++;
    }
}
```

Output

```
Enter N: 3
Perfect Numbers:
6 28 496
=== Code Execution Successful ===
```

8. Write a program to enter the marks of a student in four subjects. Then calculate the total and aggregate, display the grade obtained by the student. If the student scores an aggregate greater than 75%, then the grade is Distinction. If aggregate is $60 \geq$ and < 75 , then the grade is First Division. If aggregate is $50 \geq$ and < 60 , then the grade is Second Division. If aggregate is $40 \geq$ and < 50 , then the grade is Third Division. Else the grade is Fail.

Sample Input & Output:

Enter the marks in python: 90

Enter the marks in c programming: 91

Enter the marks in Mathematics: 92

Enter the marks in Physics: 93

Total= 366

Aggregate = 91.5

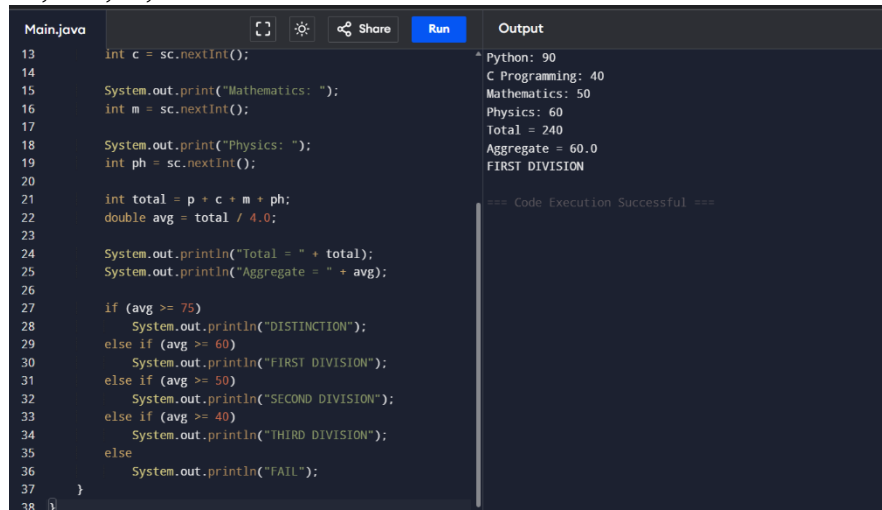
DISTINCTION

Test cases:

- a) 18, 76, 93, 65

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- b) 73,78,79,75
- c) 98,106,120,95
- d) 96,73, -85,95
- e) 78,59.8,76,79



```
Main.java
13  int c = sc.nextInt();
14
15  System.out.print("Mathematics: ");
16  int m = sc.nextInt();
17
18  System.out.print("Physics: ");
19  int ph = sc.nextInt();
20
21  int total = p + c + m + ph;
22  double avg = total / 4.0;
23
24  System.out.println("Total = " + total);
25  System.out.println("Aggregate = " + avg);
26
27  if (avg >= 75)
28      System.out.println("DISTINCTION");
29  else if (avg >= 60)
30      System.out.println("FIRST DIVISION");
31  else if (avg >= 50)
32      System.out.println("SECOND DIVISION");
33  else if (avg >= 40)
34      System.out.println("THIRD DIVISION");
35  else
36      System.out.println("FAIL");
37  }
38  }
```

Output

```
Python: 90
C Programming: 40
Mathematics: 50
Physics: 60
Total = 240
Aggregate = 60.0
FIRST DIVISION

=== Code Execution Successful ===
```

9. Write a program to read the numbers until -1 is encountered. Find the average of positive numbers and negative numbers entered by user.

Sample Input:

Enter -1 to exit...
Enter the number: 7
Enter the number: -2
Enter the number: 9
Enter the number: -8
Enter the number: -6
Enter the number: -4
Enter the number: 10
Enter the number: -1

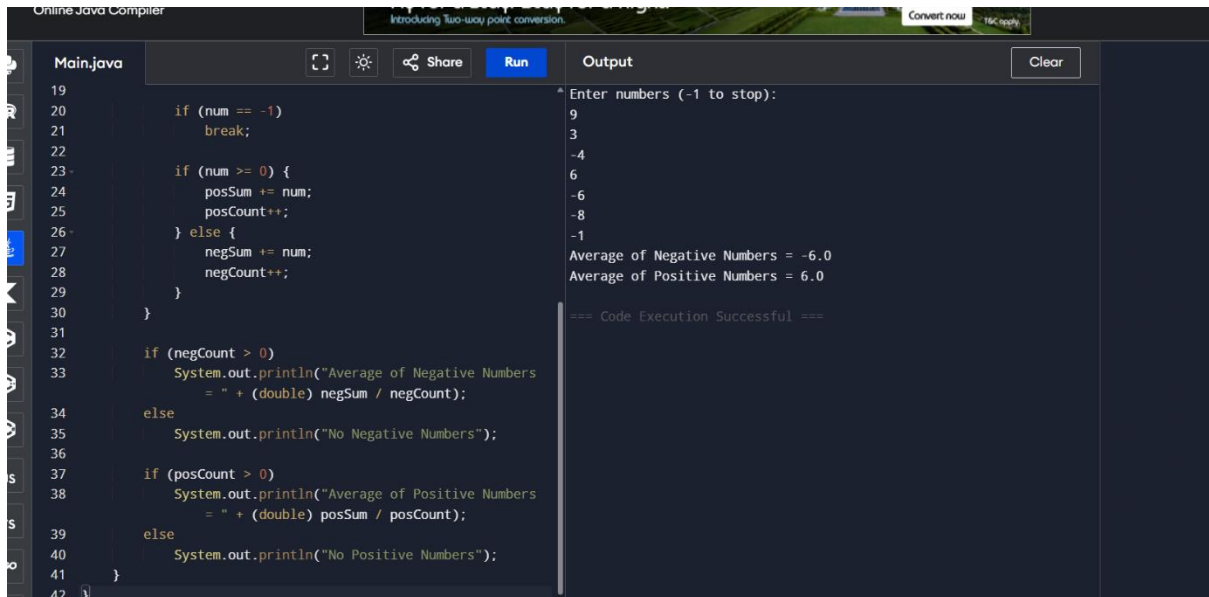
Sample Output:

The average of negative numbers is: -5.0
The average of positive numbers is : 8.666666667

Test cases:

1. -1,43, -87, -29, 1, -9
2. 73, 7-6,2,10,28,-1
3. -5, -9, -46,2,5,0
4. 9, 11, -5, 6, 0,-1
5. -1,-1,-1,-1,-1

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The screenshot shows an online Java compiler interface. On the left, the 'Main.java' file contains the following code:

```
19
20     if (num == -1)
21         break;
22
23     if (num >= 0) {
24         posSum += num;
25         posCount++;
26     } else {
27         negSum += num;
28         negCount++;
29     }
30 }
31
32 if (negCount > 0)
33     System.out.println("Average of Negative Numbers
34                        = " + (double) negSum / negCount);
35 else
36     System.out.println("No Negative Numbers");
37
38 if (posCount > 0)
39     System.out.println("Average of Positive Numbers
40                        = " + (double) posSum / posCount);
41 else
42     System.out.println("No Positive Numbers");
```

On the right, the 'Output' panel shows the execution results:

```
* Enter numbers (-1 to stop):
9
3
-4
6
-6
-8
-1
Average of Negative Numbers = -6.0
Average of Positive Numbers = 6.0
=== Code Execution Successful ===
```

10. Write a program to read a character until a * is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

Sample Input:

Enter * to exit...

Enter any character: W

Enter any character: d

Enter any character: A

Enter any character: G

Enter any character: g

Enter any character: H

Enter any character: *

Sample Output:

Total count of lower case:2

Total count of upper case:4

Total count of numbers =0

Test cases:

1. 1,7,6,9,5
2. S, Q, l, K,7, j, M
3. M, j, L, &, @, G
4. D, K, I, 6, L, *
5. *, K, A, e, 1, 8, %, *

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```
Main.java
7 Scanner sc = new Scanner(System.in);
8
9 int upper = 0, lower = 0, digit = 0;
10
11 System.out.println("Enter characters (* to stop):");
12
13 while (true) {
14     char ch = sc.next().charAt(0);
15
16     if (ch == '*')
17         break;
18
19     if (Character.isUpperCase(ch))
20         upper++;
21     else if (Character.isLowerCase(ch))
22         lower++;
23     else if (Character.isDigit(ch))
24         digit++;
25 }
26
27 System.out.println("Total Uppercase = " + upper);
28 System.out.println("Total Lowercase = " + lower);
29 System.out.println("Total Numbers = " + digit);
30 }
31 }
```

Output

```
Enter characters (* to stop):
l
i
h
i
t
h
a
k
c
h
e
r
r
\
*
Total Uppercase = 0
Total Lowercase = 13
Total Numbers = 0
=== Code Execution Successful ===
```

11. Write a program to calculate the factorial of number using recursive function.

Sample Input & Output:

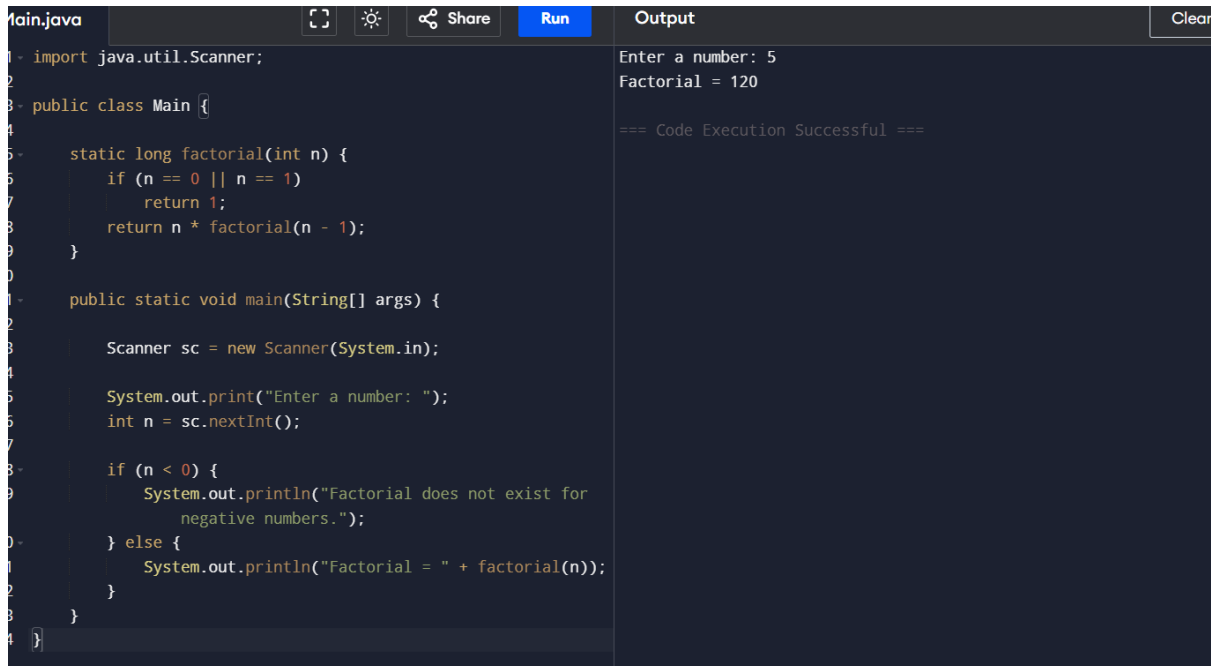
Enter the value of n: 6

Sample Input & Output:

The factorial of 6 is: 720

Test cases:

1. N = 0
2. N = -5
3. N = 1
4. N = M
5. N = %



```
Main.java
1 import java.util.Scanner;
2
3 public class Main {
4
5     static long factorial(int n) {
6         if (n == 0 || n == 1)
7             return 1;
8         return n * factorial(n - 1);
9     }
10
11     public static void main(String[] args) {
12
13         Scanner sc = new Scanner(System.in);
14
15         System.out.print("Enter a number: ");
16         int n = sc.nextInt();
17
18         if (n < 0) {
19             System.out.println("Factorial does not exist for
20             negative numbers.");
21         } else {
22             System.out.println("Factorial = " + factorial(n));
23         }
24     }
25 }
```

Output

```
Enter a number: 5
Factorial = 120
=== Code Execution Successful ===
```

12. Write a Program to Find the Nth Largest Number in a array.

Sample Input:

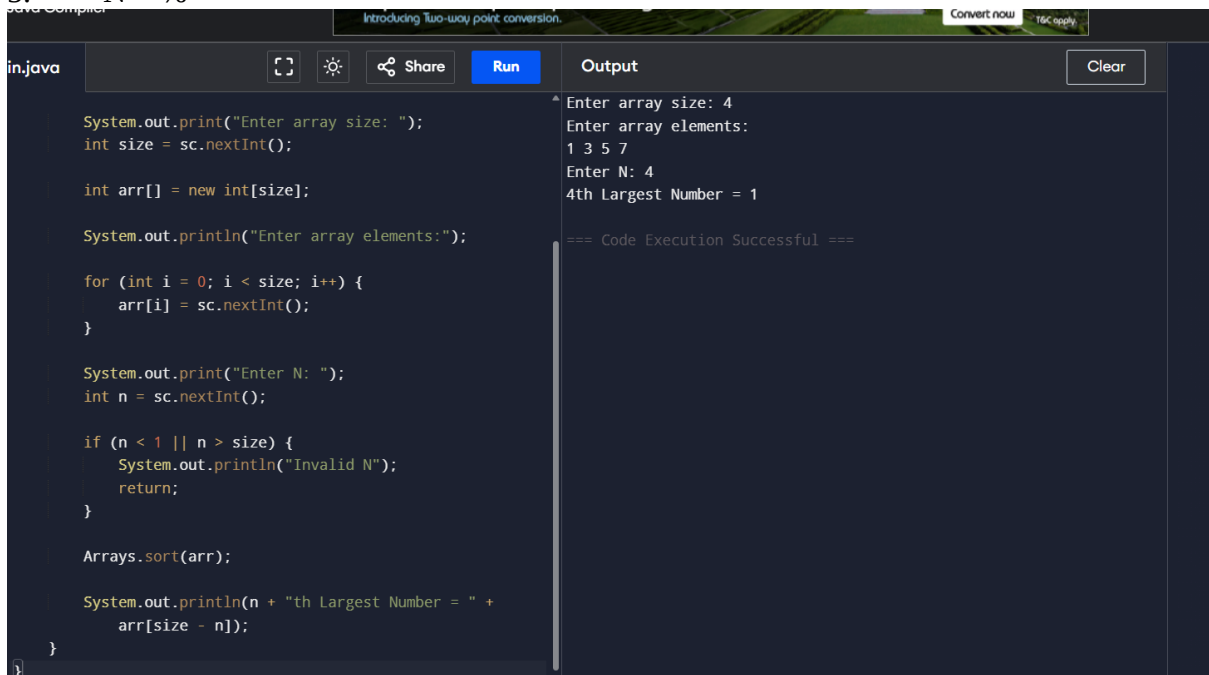
List : {14, 67, 48, 23, 5, 62}

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N = 4
Sample Output:
4th Largest number: 23

Test cases:

1. N = 0
2. N = -5
3. N = 1
4. N = M
5. N = %



The screenshot shows a Java IDE with a code editor on the left and an output window on the right. The code in the editor is as follows:

```
System.out.print("Enter array size: ");
int size = sc.nextInt();

int arr[] = new int[size];

System.out.println("Enter array elements:");

for (int i = 0; i < size; i++) {
    arr[i] = sc.nextInt();
}

System.out.print("Enter N: ");
int n = sc.nextInt();

if (n < 1 || n > size) {
    System.out.println("Invalid N");
    return;
}

Arrays.sort(arr);

System.out.println(n + "th Largest Number = " +
    arr[size - n]);
}
```

The output window shows the following text:

```
Enter array size: 4
Enter array elements:
1 3 5 7
Enter N: 4
4th Largest Number = 1
=== Code Execution Successful ===
```

13. Write a program to convert the Binary to Decimal, Octal

Sample Input:

Given Number: 1101

Sample Output:

Decimal Number: 13

Octal:15

Test cases:

1. 211
2. 11011
3. 22122
4. 111011.011
5. 1010.0101

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The screenshot shows an online Java compiler interface. The code in Main.java is as follows:

```
1- import java.util.Scanner;
2-
3- public class Main {
4-
5-     public static void main(String[] args) {
6-
7-         Scanner sc = new Scanner(System.in);
8-
9-         System.out.print("Enter Binary Number: ");
10-        String binary = sc.next();
11-
12-        int decimal = Integer.parseInt(binary, 2);
13-
14-        System.out.println("Decimal Number = " + decimal);
15-        System.out.println("Octal Number = " + Integer
16-        .toOctalString(decimal));
17-    }
18- }
```

The output shows the result of running the program with input 1101:

```
Enter Binary Number: 1101
Decimal Number = 13
Octal Number = 15

=== Code Execution Successful ===
```

14. Write a program to find the number of special characters in the given statement
Sample Input:
Given statement: Modi Birthday @ September 17, #&\$% is the wishes code for him.
Sample Output:
Number of special Characters: 5

The screenshot shows an online Java compiler interface. The code in Main.java is as follows:

```
1- import java.util.Scanner;
2-
3- public class Main {
4-
5-     public static void main(String[] args) {
6-
7-         Scanner sc = new Scanner(System.in);
8-
9-         System.out.println("Enter a statement:");
10-        String str = sc.nextLine();
11-
12-        int count = 0;
13-
14-        for (int i = 0; i < str.length(); i++) {
15-
16-            char ch = str.charAt(i);
17-
18-            if (!Character.isLetterOrDigit(ch) && ch != ' ')
19-            {
20-                count++;
21-            }
22-        }
23-
24-        System.out.println("Number of Special Characters = " + count);
25-    }
26- }
```

The output shows the result of running the program with input "likitha is a very bad girl":

```
Enter a statement:
likitha is a very bad girl
Number of Special Characters = 0

=== Code Execution Successful ===
```

15. Write a Program to Remove the Duplicate Items from a array.
Sample Input:
Enter the number of elements in array:7
Enter element1:10

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Enter element2:20

Enter element3:20

Enter element4:30

Enter element5:40

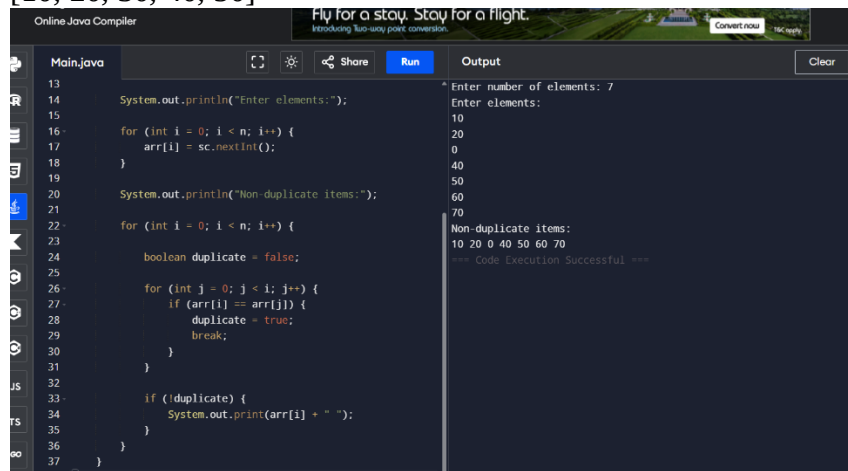
Enter element6:40

Enter element7:50

Sample Output:

Non-duplicate items:

[10, 20, 30, 40, 50]



The screenshot displays an online Java compiler interface. The code in the editor is as follows:

```
13
14 System.out.println("Enter elements:");
15
16 for (int i = 0; i < n; i++) {
17     arr[i] = sc.nextInt();
18 }
19
20 System.out.println("Non-duplicate items:");
21
22 for (int i = 0; i < n; i++) {
23
24     boolean duplicate = false;
25
26     for (int j = 0; j < i; j++) {
27         if (arr[i] == arr[j]) {
28             duplicate = true;
29             break;
30         }
31     }
32
33     if (!duplicate) {
34         System.out.print(arr[i] + " ");
35     }
36 }
37 }
```

The output window shows the following results:

```
Enter number of elements: 7
Enter elements:
10
20
0
40
50
60
70
Non-duplicate items:
10 20 0 40 50 60 70
--- Code Execution Successful ---
```